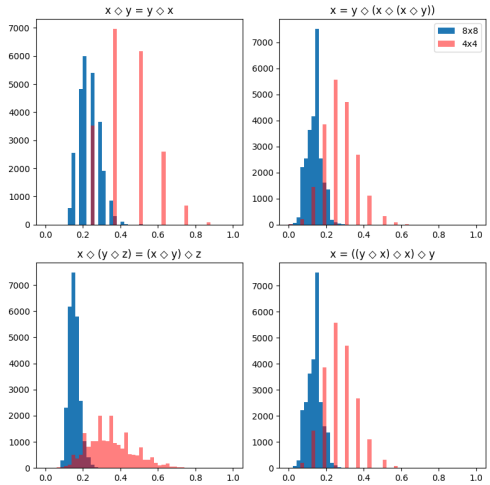


Figure 2: Stone spectra of the commutative law (top-left) and of the associative law (bottom-left) for a sampling of $n = 20\,000$ magmas of size 4 (red curve) and of $n = 24\,000$ magmas of size 8 (blue curve). The Stone spectra of the conjugate theories Eqn[63] and Eqn[271] are identical (top and bottom, right).

$$\circledast \quad : \quad S_n \times \mathbb{H}_n \longrightarrow \mathbb{H}_n$$

of the symmetry group $S_n$ on the $n$ dimensions (or directions) of the space $\mathbb{H}_n$, defined as

$$\sigma \circledast (p_1, \ldots, p_n) := (p_{\sigma(1)}, \ldots, p_{\sigma(n)})$$

for $\sigma \in S_n$ and $\mathbf{p} = (p_1, \ldots, p_n) \in \mathbb{H}_n$. Since the sampling order of the magmas $A_1, \ldots, A_n$ does not matter, it makes sense to consider the position of each equational theory

$$\mathbf{p} := (p_1, \ldots, p_n) \in \mathbb{H}_n$$

modulo the action of the symmetry group $S_n$. One obtains in this way a position in the quotient space $\mathbb{H}_n/S_n$, defined as the multiset

$$\mathbf{p}\,[S_n] \;:=\; \{\!|\, p_1 \,,\, \ldots \,,\, p_n \,|\!\}$$

We call this multiset the *Stone spectrum* of the equational theory. The Stone spectrum can be nicely visualized as a finite positive measure of weight $n$ on the interval $[0, 1]$, defined as the sum

$$\textstyle\sum_{i=1}^{n} \delta_{p_i}$$

of the Dirac distributions $\delta_{p_i}$, see Fig. 2 for an illustration.

### 5.4 Stone interference spectrum

The cartesian product of the hypercube $\mathbb{H}_n = [0, 1]^n$ with itself comes equipped with an action

$$\circledast \quad : \quad S_n \times \mathbb{H}_n \times \mathbb{H}_n \longrightarrow \mathbb{H}_{\times}\mathbb{H}_n$$

defined componentwise:

$$\sigma \circledast (\mathbf{p}, \mathbf{q}) \;:=\; (\sigma \circledast \mathbf{p}, \sigma \circledast \mathbf{q})$$

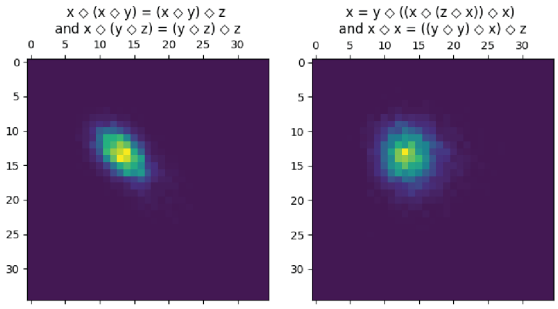


Figure 3: The Stone interference spectrum of Eqn[4400] and Eqn[4533] detects a degree of statistical dependence between the two conjugate equational theories (left) while the Stone interference spectrum of Eqn[1092] and Eqn[4092] shows full statistical independence between the two equational theories (right).

Instead of looking at the relative positions of a pair of equational theories Eqn[$j$] and Eqn[$k$] in the feature space $\mathbb{H}_n \times \mathbb{H}_n$, we may look at their relative positions in the quotient space $\mathbb{H}_n \times \mathbb{H}_n / S_n$. This is provided by the finite multiset of pairs:

$$(\mathbf{p}, \mathbf{q})\,[S_n] \;:=\; \{\!|\, (p_1, q_1) \,,\, \ldots \,,\, (p_n, q_n) \,|\!\}$$

which we call *Stone interference spectrum*. In the same way as the Stone spectrum of an equational theory, this interference spectrum of two equational theories can be described as the finite measure on the square $[0, 1]^2$ defined as the sum

$$\textstyle\sum_{i=1}^{n} \sum_{j=1}^{n} \delta_{(p_i, q_j)}$$

where $\delta_{(p,q)}$ denotes the Dirac distribution of $(p, q) \in [0, 1]^2$. This representation enables one to visualize the Stone interference spectrum in an instructive way, see Fig. 2 for two illustrations.

## 6 The latent space of equational theories

### 6.1 Building the latent space using a PCA dimensionality reduction

We apply a principal component analysis (PCA) method in order to derive a 3-dimensional representation of the 4694 equational theories from thr positions in the $n$-dimensional feature space $\mathbb{H}_n$. We thus start by computing the center of gravity $\mu \in \mathbb{H}_n$ of the 4694 vertices in $\mathbb{H}_n$. We subtract $\mu$ from each column of the matrix $R$ defined in (1), in order to obtain a matrix $X = R - \mu$ with zero-mean rows. The PCA method then diagonalizes the covariance matrix $C_X = \frac{1}{n} X X^T$, see (Shlens, 2014; Hanchi et al., 2025) for details.

We keep the first three principal components or dimensions (noted $X, Y, Z$) of the row-space which we call the *latent space of equational theories*. This